STATIC ROUTING & FLOATING STATIC ROUTES

Purpose

In this lab, we dive into Layer 3 concepts, where we'll tackle basic static routes, multi-hop routing logic, default routes, as well as floating static routes for backup paths.

Topology

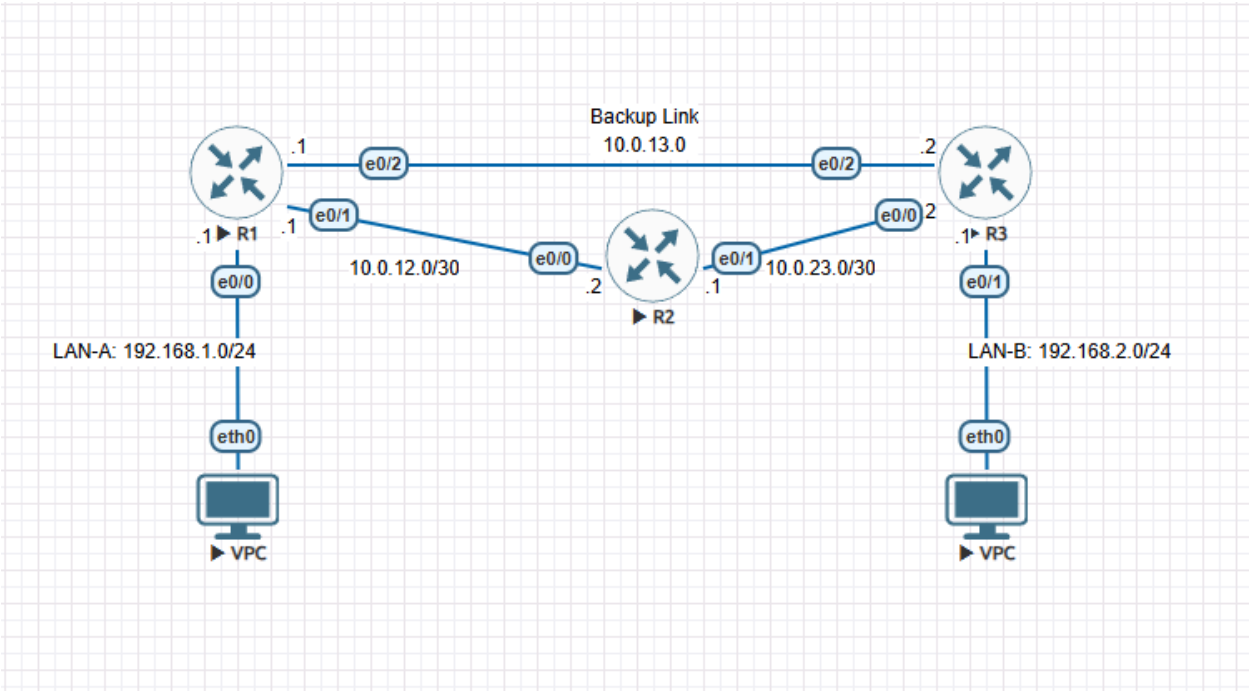

Figure 1: Topology

Requirements

Part A - Basic Static Routing

1. Configure all router interfaces for all 3 routers
2. Configure PC IP addresses
3. Add static routes to and from LAN A and LAN B across all 3 routers

Part B - Default Routes

1. Simulate an edge router on R1 and R3 by applying default routes

Part C - Floating Static Routes (Backup Path)

1. Set up a floating static route utilizing the backup link between R1 and R3
2. Simulate a link failure on the original path (R1 ⟷ R2 ⟷ R3) to test backup link.

Tasks

Part A

First we configure the interfaces on each of the routers

R1

```
R1(config)#int e0/0
R1(config-if)#description LAN-A Gateway
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#int e0/1
R1(config-if)#description Link to R2
R1(config-if)#ip address 10.0.12.1 255.255.255.252
R1(config-if)#no shut
R1(config-if)#int e0/2
R1(config-if)#description Backup Link to R3
R1(config-if)#ip address 10.0.13.1 255.255.255.252
```

R2

```
R2(config)#interface e0/0
R2(config-if)#description Link to R1
R2(config-if)#ip address 10.0.12.2 255.255.255.252
R2(config-if)#no shutdown
R2(config-if)#int e0/1
R2(config-if)#description Link to R3
R2(config-if)#ip address 10.0.23.1 255.255.255.252
R2(config-if)#no shut
```

R3

```
R3(config)#interface e0/0
R3(config-if)#description Link to R2
R3(config-if)#ip add 10.0.23.2 255.255.255.252
R3(config-if)#no shut
R3(config)#int eth0/1
R3(config-if)#description LAN-B Gateway
R3(config-if)#ip address 192.168.2.1 255.255.255.0
R3(config-if)#no shut
R3(config-if)#int eth0/2
R3(config-if)#description Backup Link to R1
R3(config-if)#ip address 10.0.13.2 255.255.255.252
R3(config-if)#end
```

Once that is done, we then give our PCs IP addresses

VPC - A

```
VPCS> ip 192.168.1.2/24 192.168.1.1
Checking for duplicate address...
VPCS : 192.168.1.2 255.255.255.0 gateway 192.168.1.1
```

VPC - B

```
VPCS> ip 192.168.2.2/24 192.168.2.1
Checking for duplicate address...
VPCS : 192.168.2.2 255.255.255.0 gateway 192.168.2.1
```

Then we can go ahead and add our static routes on the routers between LAN-A and LAN-B

R1

```
R1(config)#ip route 192.168.2.0 255.255.255.0 10.0.12.2
```

R2

```
R2(config)#ip route 192.168.2.0 255.255.255.0 10.0.23.2
R2(config)#ip route 192.168.1.0 255.255.255.0 10.0.12.1
```

R3

```
R1(config)#ip route 192.168.2.0 255.255.255.0 10.0.12.2
```

Once the routes have been added, we can try to ping PC-B from PC-A.

Part B - Default routes

Now for part B, we'll simulate "edge" routers on R1 and R3 by using default routes.

R1

```
R1(config)#ip route 0.0.0.0 0.0.0.0 10.0.12.2
```

R3

```
R3(config)#ip route 0.0.0.0 0.0.0.0 10.0.23.1
```

Part C - Floating Static Routes (Backup Path)

Now we'll add additional routes on R1 and R3 as a backup in case of link failure between R1, R2 and R3.

R1

```
R1(config)#ip route 192.168.2.0 255.255.255.0 10.0.13.2 200
R1(config)#int eth0/2
R1(config-if)#no shut
```

R3

```
R3(config)#ip route 192.168.1.0 255.255.255.0 10.0.13.1 200
R3(config)#int eth 0/2
R3(config-if)#no shut
```

To simulate a link failure, we'll shut down Eth0/0 on R3, and Eth0/1 on R1

R3

```
R3(config)#int eth0/0
R3(config-if)#shut
```

R1

```
R1(config)#int eth 0/1
R1(config-if)#shut
```

Verification

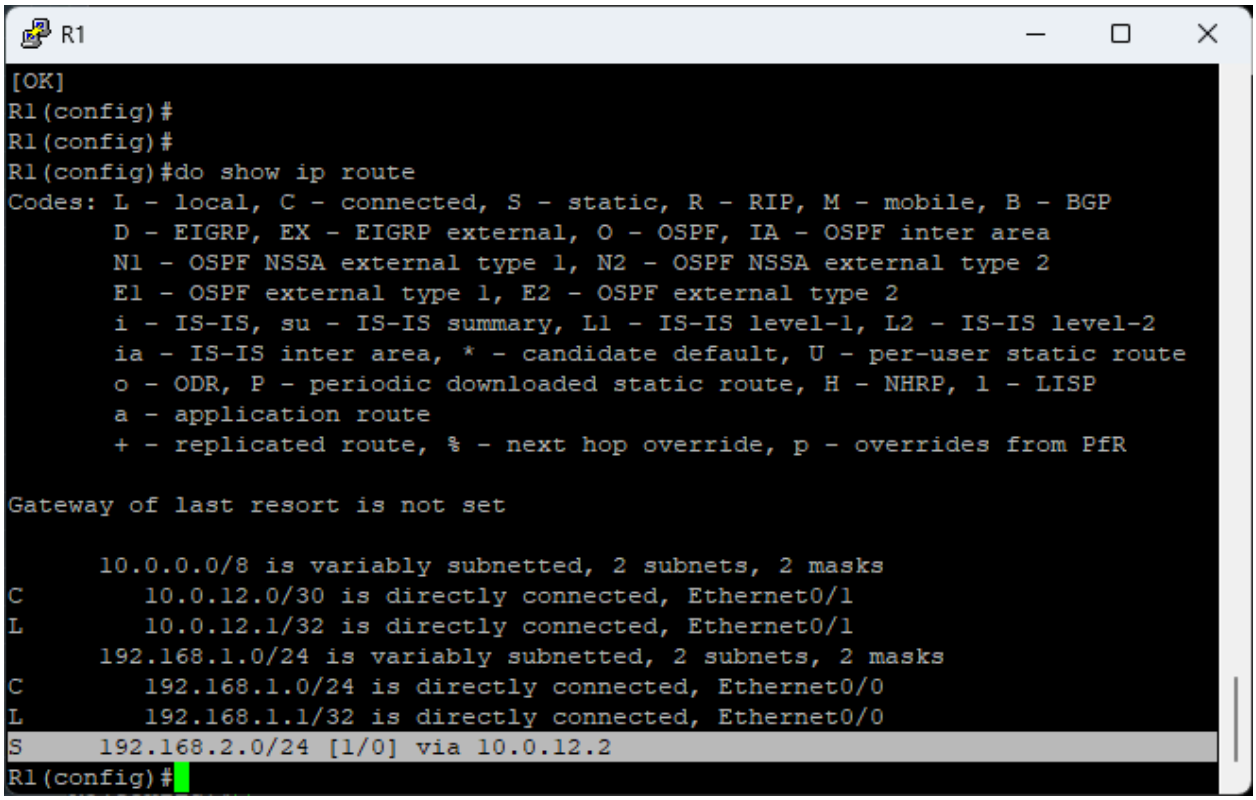

Figure 2: `show ip route` output on R1

```
R2
R2(config)#do write
Building configuration...
[OK]
R2(config)#do show IP route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C        10.0.12.0/30 is directly connected, Ethernet0/0
L        10.0.12.2/32 is directly connected, Ethernet0/0
C        10.0.23.0/30 is directly connected, Ethernet0/1
L        10.0.23.1/32 is directly connected, Ethernet0/1
S     192.168.1.0/24 [1/0] via 10.0.12.1
S     192.168.2.0/24 [1/0] via 10.0.23.2
R2(config)#
```

Figure 3:`show ip route` output on R2

```
R3
R3(config)#do write
Building configuration...
[OK]
R3(config)#do show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        10.0.23.0/30 is directly connected, Ethernet0/0
L        10.0.23.2/32 is directly connected, Ethernet0/0
S     192.168.1.0/24 [1/0] via 10.0.23.1
      192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.2.0/24 is directly connected, Ethernet0/1
L        192.168.2.1/32 is directly connected, Ethernet0/1
R3(config)#
```

Figure 4: `show ip route` output on R3

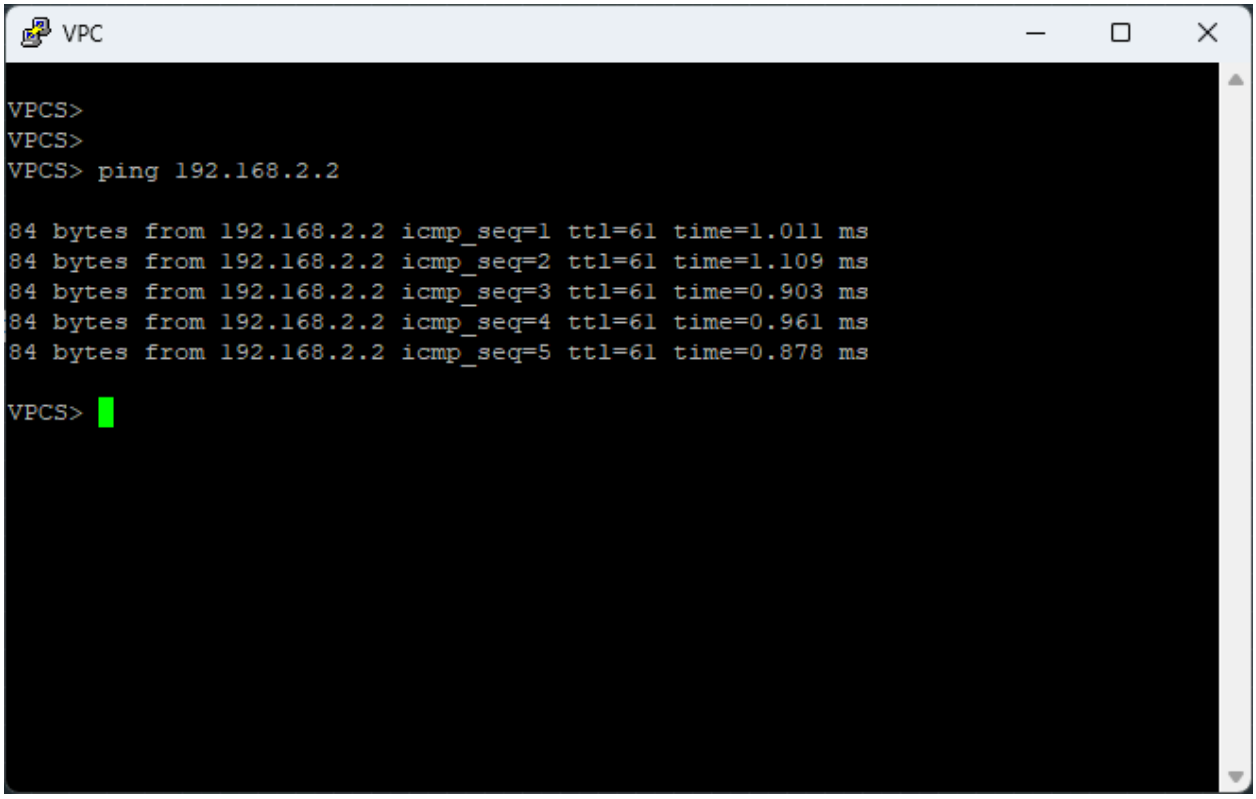

Figure 5: Ping to PC-B from PC-A

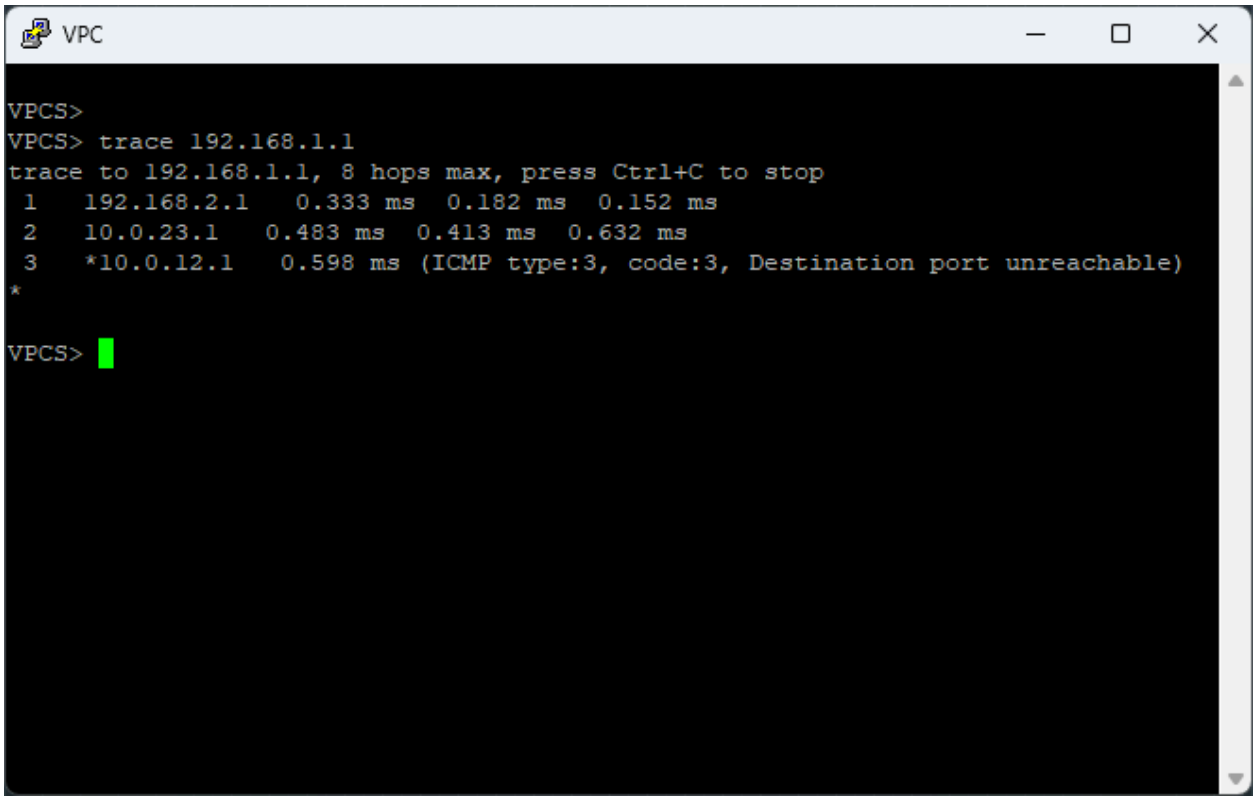

Figure 6: Traceroute to PC-A from PC-B (Original Path)

```
R3
Building configuration...
[OK]
R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is 10.0.23.1 to network 0.0.0.0

S*    0.0.0.0/0 [1/0] via 10.0.23.1
      10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        10.0.23.0/30 is directly connected, Ethernet0/0
L        10.0.23.2/32 is directly connected, Ethernet0/0
S     192.168.1.0/24 [1/0] via 10.0.23.1
      192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.2.0/24 is directly connected, Ethernet0/1
L        192.168.2.1/32 is directly connected, Ethernet0/1
R3#
```

Figure 7: `show ip route` output on R3 (Default route added)

```
R3
*Jan 22 17:20:02.627: %SYS-5-CONFIG_I: Configured from console by console
R3#
R3#
R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        10.0.13.0/30 is directly connected, Ethernet0/2
L        10.0.13.2/32 is directly connected, Ethernet0/2
S     192.168.1.0/24 [200/0] via 10.0.13.1
      192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.2.0/24 is directly connected, Ethernet0/1
L        192.168.2.1/32 is directly connected, Ethernet0/1
R3#
```

Figure 8: `show ip route` output on R3 (Backup floating route)

```
R1#
R1#
*Jan 22 17:27:01.807: %SYS-5-CONFIG_I: Configured from console by console
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        10.0.13.0/30 is directly connected, Ethernet0/2
L        10.0.13.1/32 is directly connected, Ethernet0/2
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.1.0/24 is directly connected, Ethernet0/0
L        192.168.1.1/32 is directly connected, Ethernet0/0
S     192.168.2.0/24 [200/0] via 10.0.13.2
R1#
```

Figure 9: `show ip route` output on R1 (Backup floating route)

```
VPCS>
VPCS>
VPCS> trace 192.168.1.1
trace to 192.168.1.1, 8 hops max, press Ctrl+C to stop
 1   192.168.2.1   0.260 ms  0.256 ms  0.180 ms
 2   *10.0.13.1   0.458 ms (ICMP type:3, code:3, Destination port unreachable)
*

VPCS>
```

Figure 10: Traceroute to PC-A from PC-B (Backup floating route)

Conclusion

In this lab, we went through the basics of static routing. When using static routing, we define the destination address with its subnet mask, as well as the next hop to that address from the router. We also observed the use of a default route. The default route acts as a “catch all” route, meaning all addresses that are not defined on the routing table will be sent to the next hop defined on the default route. Remember that when a router receives a packet to be routed, it checks against all the entries in the routing table for one with the highest prefix match. If none is found it’ll send the packet via the default route, also known as the gateway of last resort.

We also learned about the concept of floating routes. Floating routes are typically used as a backup in case of link failures. In order to set a floating static route, we need to configure our static route with an administrative distance that is higher than 1 (1 is the default administrative distance of a static route). After the highest prefix match, the router will prioritize the route with the lowest administrative distance. Therefore, by setting our administrative distance to 200 for the same route, the router will prioritize the route with the AD of 1 instead of 200. Also note that when both the routes are configured, only the active route (AD = 1) will be displayed on the routing table. When we simulate a link failure on the original path, the floating route takes precedence as the original route is no longer valid / next hop is no longer reachable. We also observed the paths the packets took before and after the link failure between LAN-A and LAN-B using traceroutes.